



Memphis Zoo Custodial System

Operational design, screen walkthrough, and cost comparison

A practical view of how the custodial system connects daily area ownership, scan-based work confirmation, guest issue reporting, events, reminders, messaging, and Memphis AI into one working operations flow.

01 See the work

Managers can see scan activity, completion status, guest issues, and schedules from the screens they already use.

02 Guide the day

Custodians get simple phone workflows: assigned areas, scan prompts, event notices, due soon reminders, and issue notifications.

03 Keep proof simple

NFC/QR activity creates a time-stamped trail without asking employees to manage spreadsheets or paperwork.

04 Control cost

The design uses low-cost phones, NFC tags, kiosk licenses, GitHub Pages, Supabase, Render, and zoo Wi-Fi instead of broad commercial software licensing.

Important framing

The system does not replace manager inspections. It tells managers when and where cleaned areas are ready to inspect, so checks can happen before guest traffic re-soils the area.

Document style

This version uses the normal print/PC page layout and full application screenshots. It is not a phone-friendly image deck.

How the System Works

The core idea is simple: a scan or report becomes a visible operations signal, and the right person gets notified.

1. Area ownership

Schedules assign custodians to restroom, exhibit, and service areas for the day.

2. Work starts at the location

The custodian uses the phone workflow and NFC/QR scan points to identify the real location.

3. Local-first activity

If Wi-Fi is weak, the phone keeps working locally and queues the action.

4. Sync and visibility

When connection returns, queued activity syncs to the backend and dashboard views refresh.

5. Messenger notifications

Guest issues, event notices, due soon locations, and overdue locations create Memphis Messenger alerts.

6. Manager inspection

Managers use the screens to know what is ready to inspect; the software supports the inspection, it does not replace it.

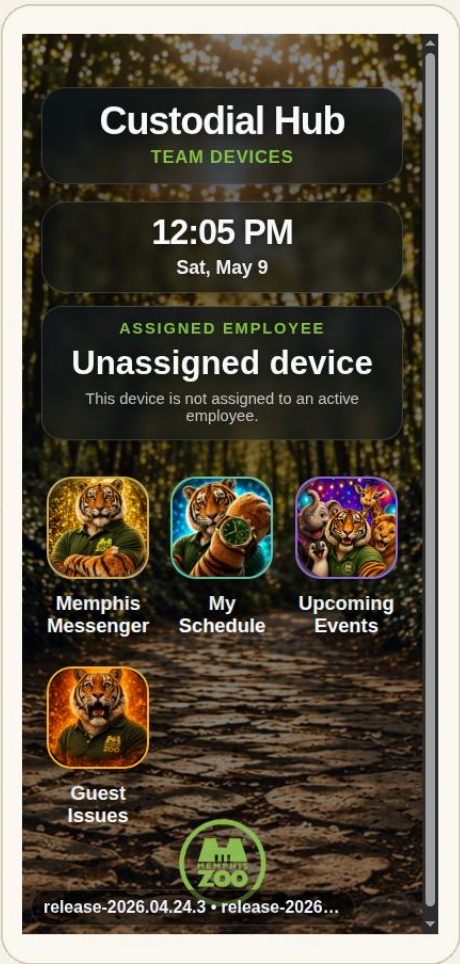
The system is strongest because the pieces are connected. Commercial tools often handle inspections, work orders, QR requests, or schedules separately. This design ties the custodial workflow together around the actual zoo day: staffing, locations, guests, events, scans, messages, and manager follow-up.

WHAT CHANGES OPERATIONALLY

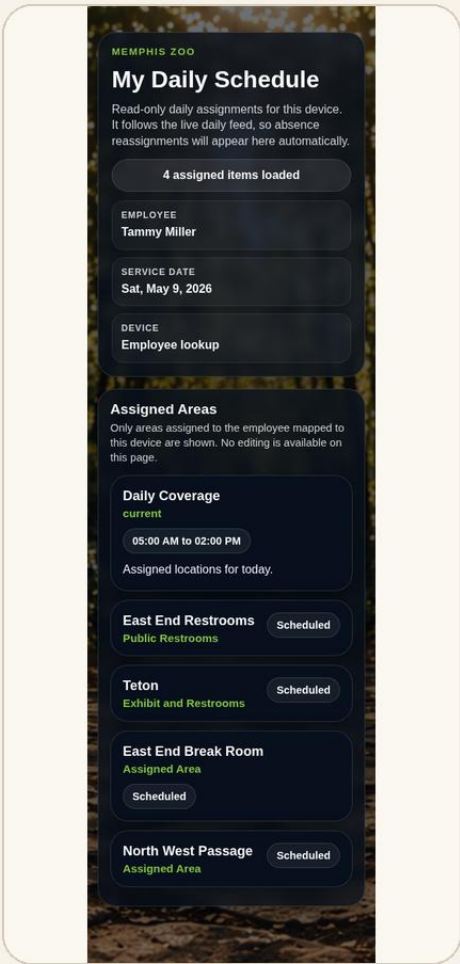
- Less guessing about which areas are ready for inspection.
- Less dependence on radio calls and memory when mornings get busy.
- Clearer response path when a guest reports a cleanliness issue.
- Event information reaches the people assigned to the affected area.
- Due soon and overdue reminders help prevent quiet misses.

Custodian Phone Views

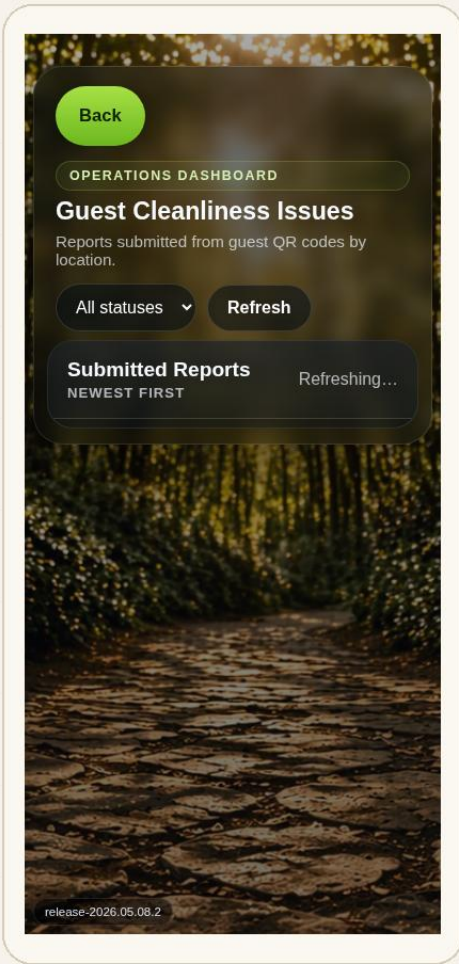
The phone side is intentionally simple: open the hub, choose the task, scan or report, and keep moving.



Employee Hub



Tammy Miller — My Schedule



Guest Issues View

HOW THE PHONE WORKFLOW SHOULD FEEL

- Custodians do not need to understand the backend. They see their schedule, assigned areas, messages, events, and guest issues.
- Tammy’s schedule shows assigned locations such as Teton as Exhibit and Restrooms when both workflows apply.
- A guest QR issue becomes a Memphis Messenger notification so the assigned custodian and operations contact can respond.
- The same notification pattern supports events in the employee’s area, due soon reminders, and overdue location reminders.
- The workflow is built for zoo Wi-Fi. If connection drops, work can continue locally and sync when service returns.

Design target

The phone screens are for quick action in the field. The PC screens are for planning, visibility, and follow-up.

PC view for daily visibility: cleaned areas, scan activity, open issues, and inspection readiness.

Back



OPERATIONS DASHBOARD

Custodial Tracking System

TODAY
1:37 PM
Sat, May 9, 2026
CURRENT ATTENDANCE
8,124

Restrooms Overdue
1

Restrooms Not Cleaned
0

Open Tickets
2

Exhibits Overdue
1

Exhibits Not Cleaned
0

Restroom Status Report

LIVE OPERATIONAL STATE

LOCATION	STATUS	TIME	TIME TO CLEAN	SERVICES PERFORMED	EMPLOYEE NAME
Teton Trek Restrooms	OVERDUE	Yesterday 4:58 PM	22 min	Full cleaning and inspection completed	Tammy Miller
China Restrooms	DUE SOON	Today 8:16 AM	16 min	Floors swept and mopped, Soap restocked, Trash emptied and receptacles cleaned	Marcus Reed
Aquarium Restrooms	OKAY	Today 9:42 AM	18 min	Toilets and urinals cleaned, Sinks and counters cleaned, Paper towels restocked, Final inspection completed before scanning the location tag	Tammy Miller
Front Entrance Restrooms	OKAY	Today 10:08 AM	14 min	Mirrors and stainless steel surfaces cleaned and polished, Toilet tissue restocked, Soap restocked	Dana Cole

Exhibit Status Report

LIVE OPERATIONAL STATE

LOCATION	STATUS	TIME	TIME TO CLEAN	SERVICES PERFORMED	EMPLOYEE NAME
Cat Country	OVERDUE	Yesterday 3:40 PM	27 min	Floors swept or vacuumed, including corners, Walls, signage, switch plates, and baseboards spot-cleaned	Riley Brooks
Aquarium Exhibit	DUE SOON	Today 7:35 AM	19 min	Entrance and exit doors cleaned, Exterior glass cleaned	Jordan Lee
Primate Canyon	OKAY	Today 10:21 AM	21 min	Stairs swept, including around railings, Interior glass cleaned	Avery Smith
Teton Trek Exhibit	OKAY	Today 9:55 AM	24 min	Floors swept or vacuumed, including corners, Interior glass cleaned, Final inspection completed before scanning the location tag	Tammy Miller

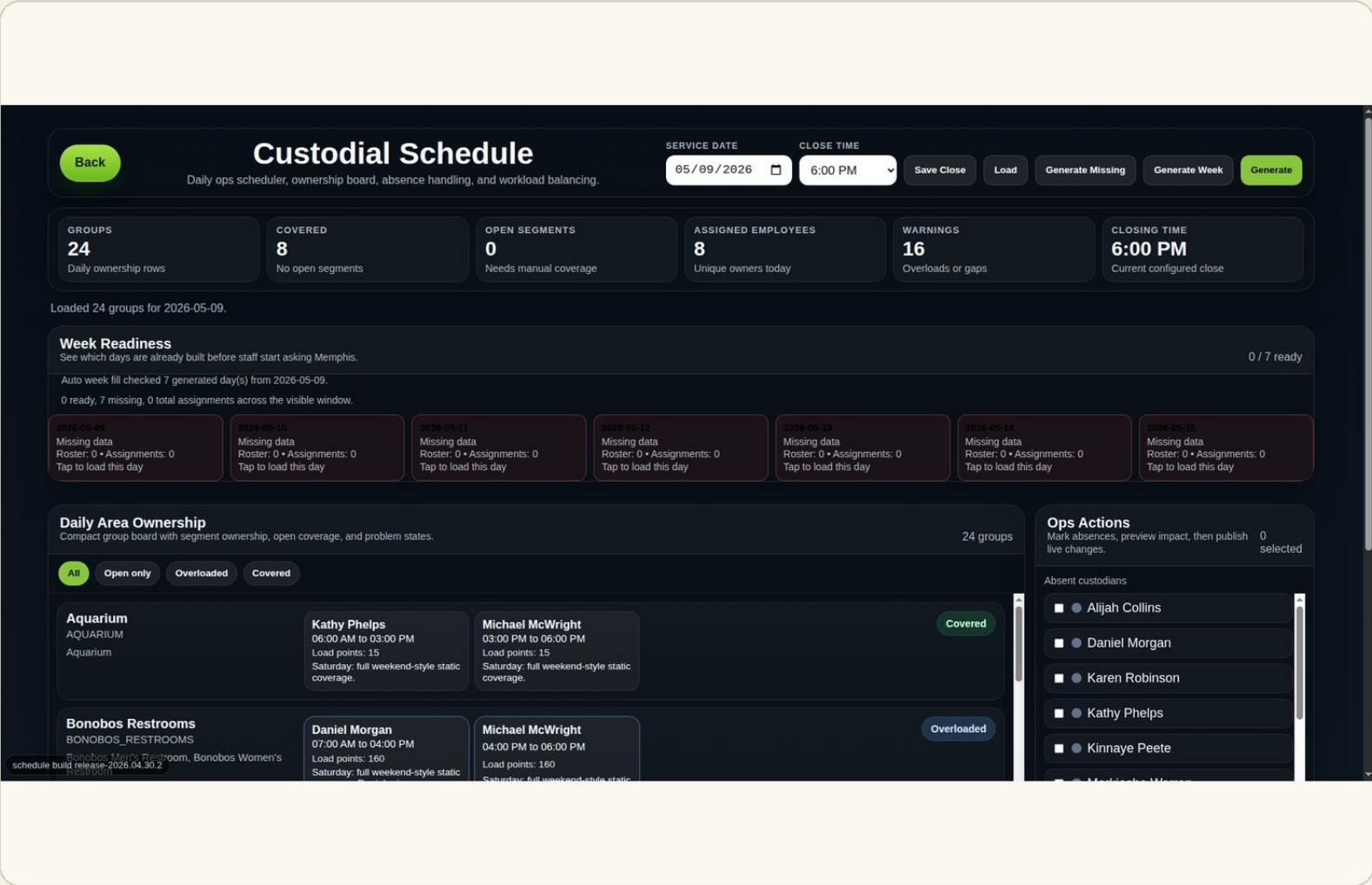
release-2026.04.25.3

WHAT THIS SCREEN SHOWS

- Shows the manager-facing control view rather than asking someone to track restroom/exhibit status manually.
- Refresh behavior keeps the screen current while the day is moving. The screenshot is used as a screen example, not a claim that rollout is complete everywhere.
- The goal is not to replace inspection; it helps managers decide where inspection attention should go next.
- The value is quick visibility: what is active, what needs follow-up, and what is ready for a manager to check.

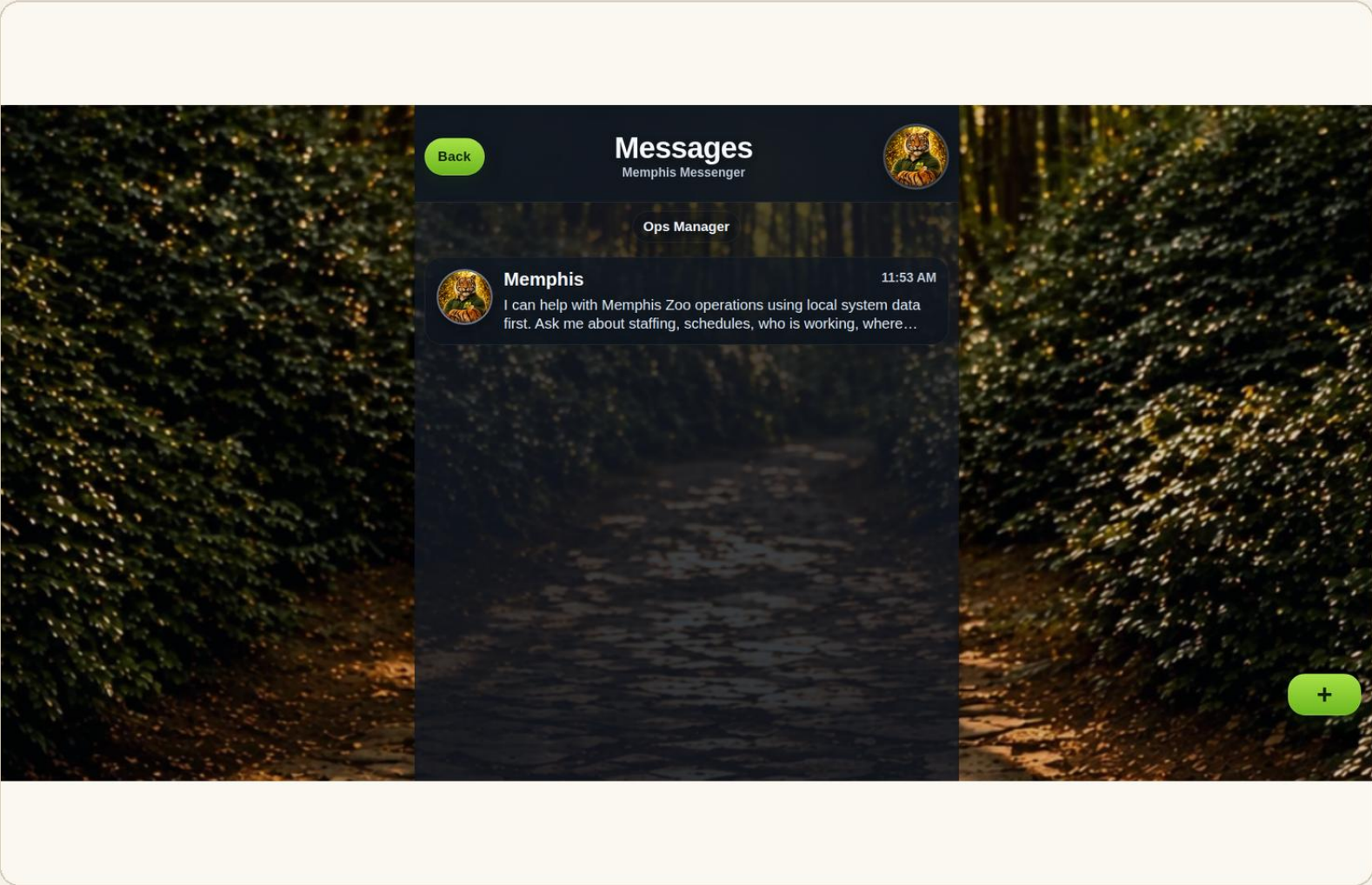
Daily Schedule and Absence Reassignment05

PC view for area ownership, close time, absence handling, and coverage planning.



WHAT THIS SCREEN SHOWS

- Daily area ownership connects employees to location groups before the day begins.
- Absence tools preview and apply reassignment instead of leaving coverage decisions buried in notes or memory.
- Close-time and workload settings give the system context for reminders and coverage expectations.
- This keeps coverage decisions tied to the same system that handles scan activity and reminders.

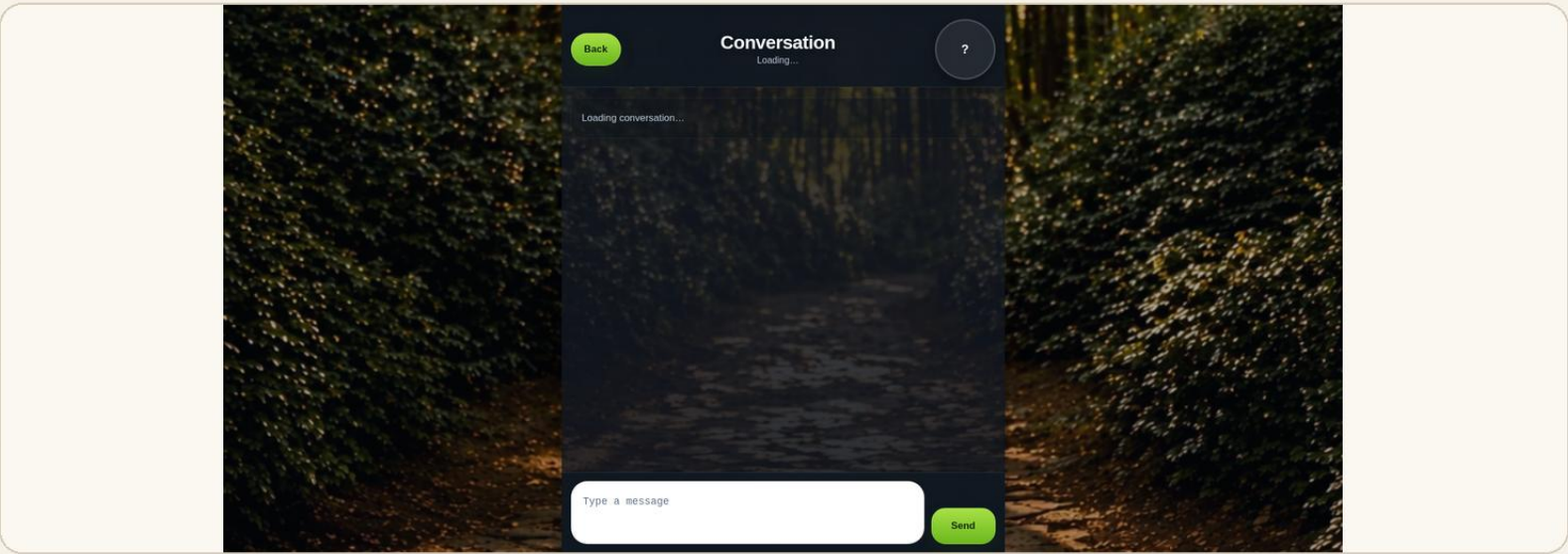
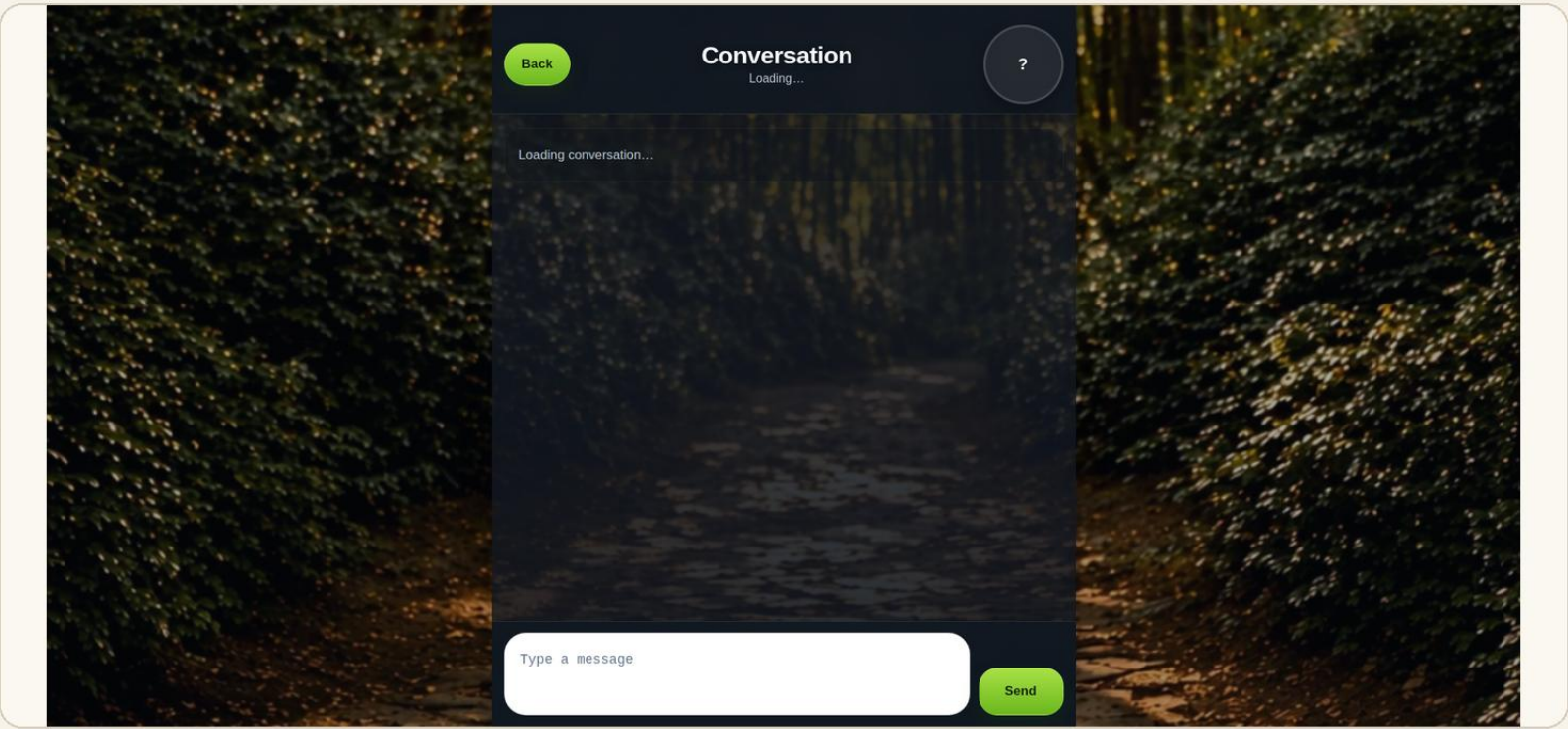


WHAT THIS SCREEN SHOWS

- Messenger is the notification path for operational prompts, not just general chat.
- Guest issues can trigger Memphis Messenger notifications to the assigned custodian and operations contact.
- Events in an assigned area, due soon locations, and overdue locations follow the same notification concept.
- This gives staff a direct place to receive prompts and ask follow-up questions during the day.

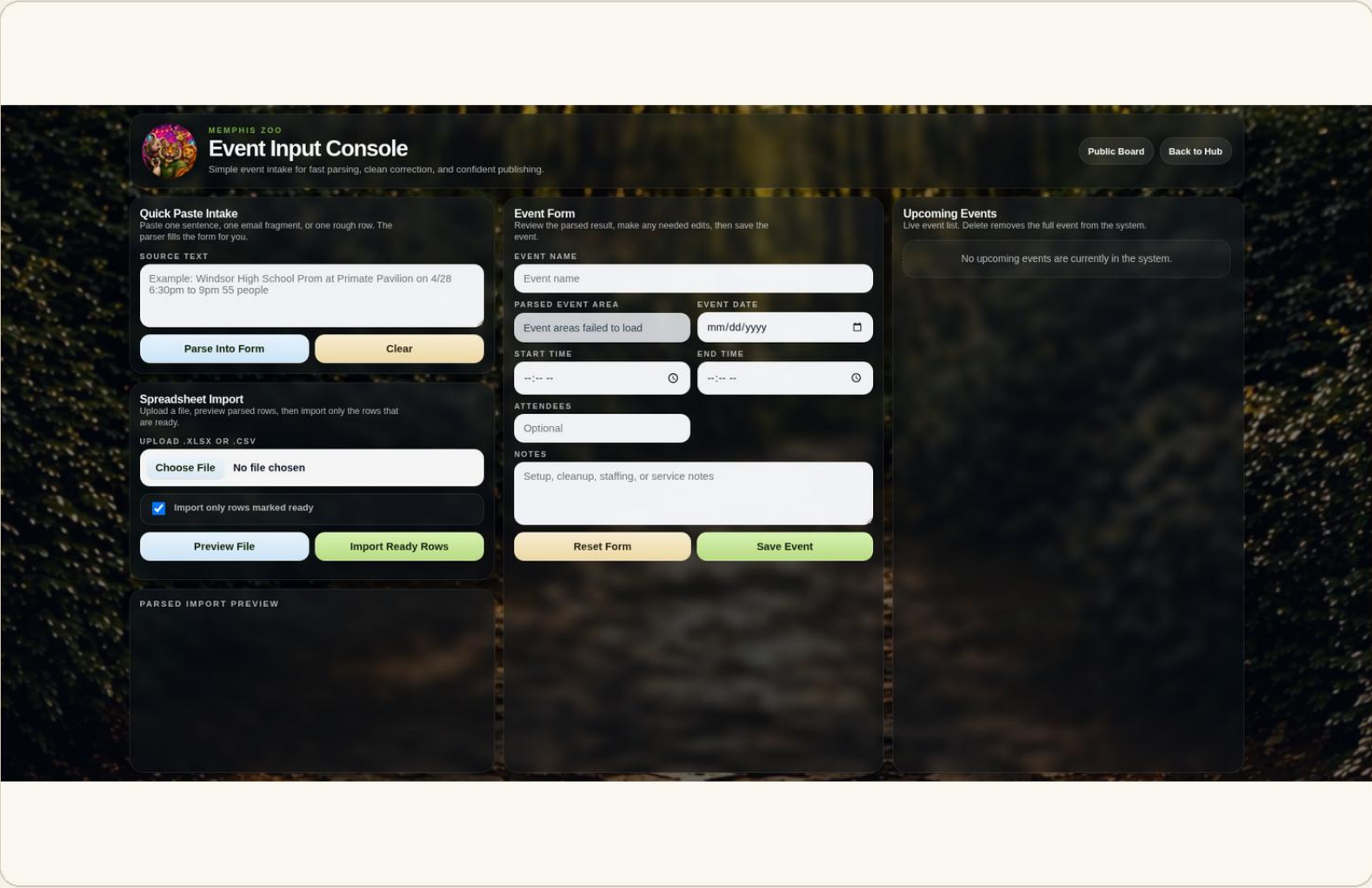
Memphis AI and Messenger

Messaging is not separate from operations; it is how the system turns data into timely reminders and answers.



WHAT MEMPHIS ADDS

- Answers operational questions from system data: staffing, schedules, area coverage, tickets, scans, events, attendance, and contacts.
- Keeps routine answers inside Memphis Messenger instead of forcing a manager to hunt through several screens.
- Supports notification workflows: guest issue reported, event in assigned area, due soon location, and overdue location.
- The live system can answer based on real assignments and current data, which makes it useful during the actual workday.

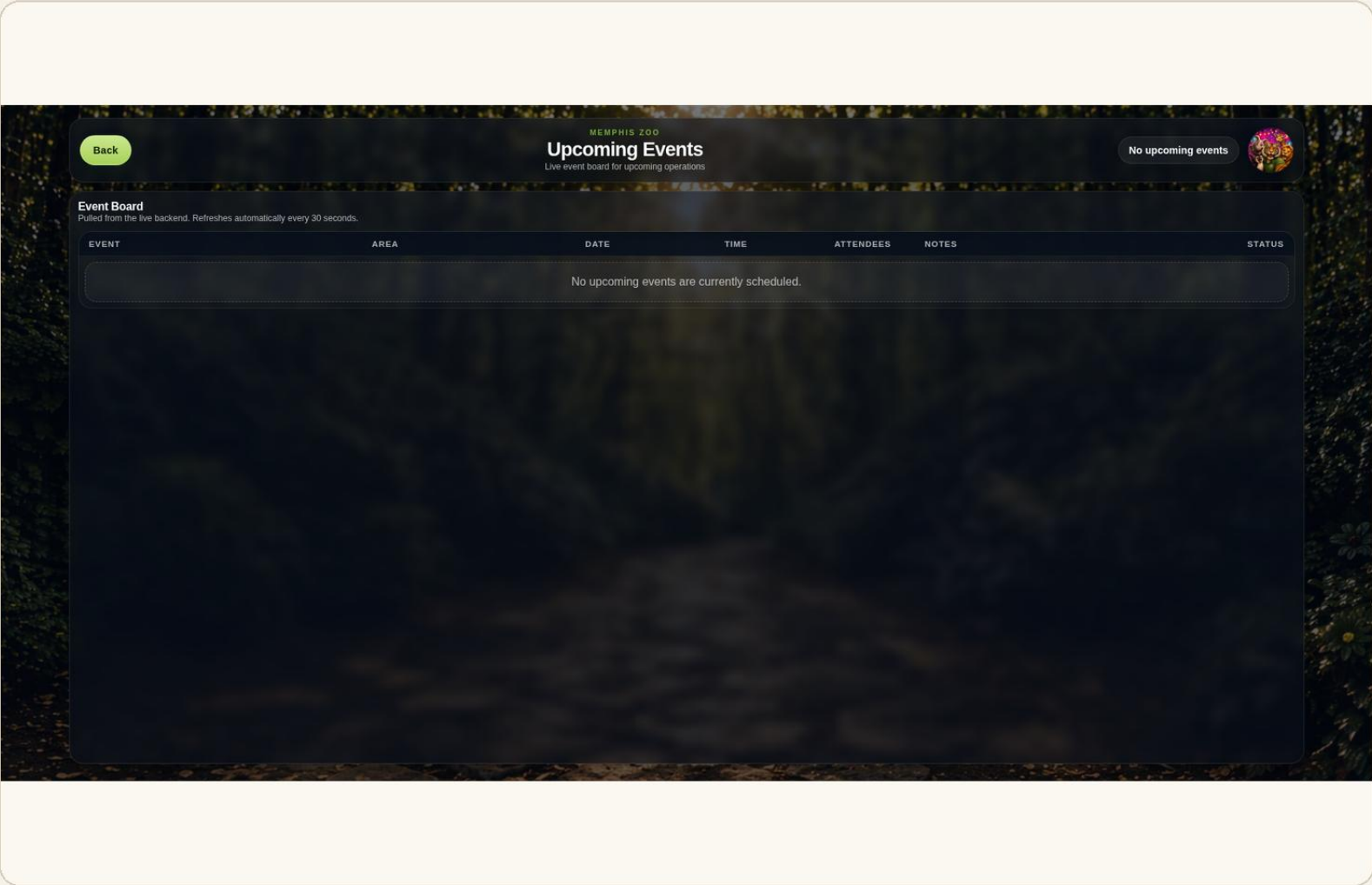


WHAT THIS SCREEN SHOWS

- Events can be entered from a quick paste, spreadsheet import, or structured form.
- The event area matters because assigned custodians can be notified when something affects their zone.
- The public board and operational screens can pull from the same event data instead of separate lists.
- This supports “events in your area today” notifications through Memphis Messenger.

Upcoming Events Board

PC view for the clean event board format.

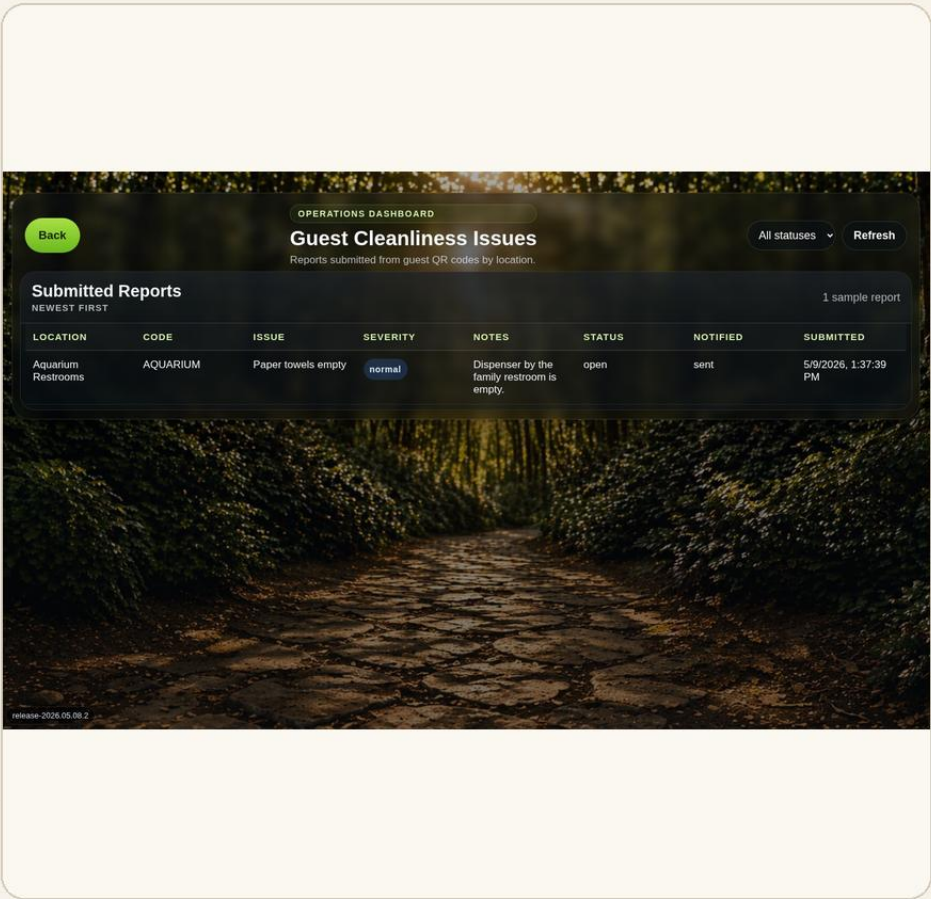
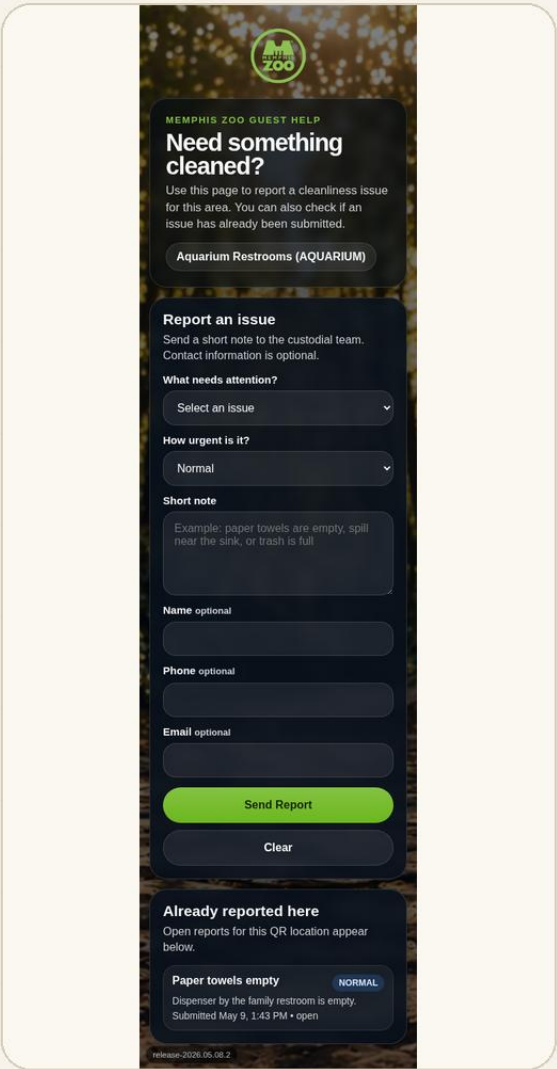


WHAT THIS SCREEN SHOWS

- Shows the public/manager-readable event board layout.
- Refresh behavior keeps event information aligned with backend data.
- When events exist, they can be tied back to areas so the right custodial coverage receives the right prompt.
- The empty state is useful because it makes clear when no event is currently scheduled.

Guest Issue Reporting

A guest report should become an operations signal, not a loose comment that someone has to manually route.



Operations-side guest issue view

Guest-facing QR landing page with sample issue

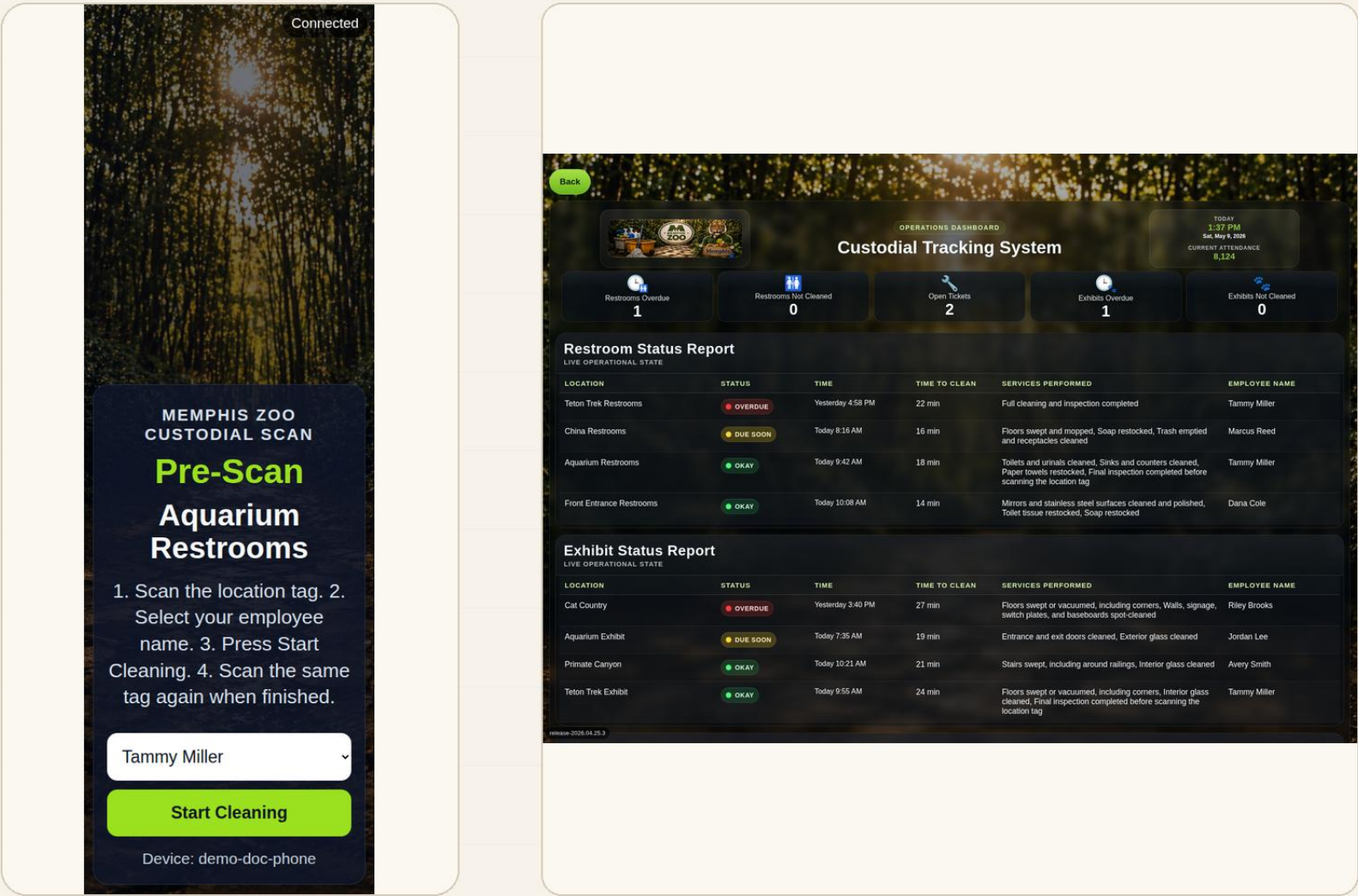
NOTIFICATION PATH

- Guest scans the location QR code and sees a simple landing/report page for that area.
- The sample issue appears on the guest page and on the operations-side issue screen so the path is easy to understand.
- Memphis Messenger notifies the assigned custodian and operations contact, following the same pattern as event, due soon, and overdue notifications.
- Managers can review submitted issues from the operations screen and close the loop after follow-up.

Why this matters

Guest-facing QR reporting turns a complaint into a routed task with a location, timestamp, and follow-up path.

The scan workflow gives custodians clear instructions and gives managers a PC view of what is okay, due soon, or overdue.



Custodian scan instructions

Manager / ops dashboard with realistic examples

WHAT THE WORKFLOW PROVES

- Custodians scan the location tag, select their name, start cleaning, and scan again when finished.
- The dashboard converts scan activity into simple visibility: OKAY, DUE SOON, OVERDUE, and open tickets.
- Restroom and exhibit examples appear together, so managers can see the full custodial picture from a PC.
- This does not replace inspection; it tells managers when and where cleaned areas are ready to inspect.

Operational value

The phone screen guides the field workflow. The PC screen turns that field activity into inspection readiness and follow-up visibility.

Notes and submit

- Restroom checklist includes toilets/urinals, sinks/counters, mirrors, floors, trash, paper products, soap, odors, and final inspection.
- The same form captures fixture-level maintenance issues, out-of-order needs, and notes before submission.
- The final submission creates the record managers use for follow-up and inspection readiness.

Notes and submit

- Exhibit checklist includes floors, glass, stairs, railings, doors, signage, baseboards, cobwebs, trash, and final inspection.
- Maintenance issues and notes are captured before submission, just like the restroom workflow.
- The final submission creates the record managers use for follow-up and inspection readiness.

Actual Project Cost and Monthly Cost14

The implementation budget is based on the attached equipment table. The phones are included as a real one-time equipment cost; there is still no phone carrier plan and no current monthly software cost.

ACTUAL EQUIPMENT PURCHASES

Item	Qty	Unit Cost	Subtotal
Moto G Phones	10	\$39.99	\$399.90
USB Charging Cords	2	\$7.59	\$15.18
Charging Station	1	\$31.98	\$31.98
Phone Holsters	10	\$12.99	\$129.90
NFC Tags	4	\$13.99	\$55.96
NFC Reader	1	\$37.99	\$37.99

Total Project Cost	\$670.91
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Monthly cost clarification

The equipment above is the real implementation purchase total. The phones do not require a carrier subscription because they run on zoo Wi-Fi. Current monthly software cost remains \$0/month for the frontend/backend services

CURRENT SERVICE COST

GitHub Pages	\$0/month current	Hosts the static frontend pages.
Supabase	\$0/month current	Current backend data layer remains on the free tier unless scale, backups, or policy needs change.
Render	\$0/month current	Current API/backend hosting remains on the free tier unless reliability or scale needs change.

EXPANSION MODEL

- Adding more locations mainly means adding more NFC/QR points and configuration records, not buying a new software package.
- Adding more phones means additional low-cost hardware and accessories, not a carrier bill when the phones stay on zoo Wi-Fi.
- The realistic ongoing cost is replacing broken phones, damaged/missing NFC tags, charging accessories, or optional cloud upgrades.
- The system avoids paying for unused CMMS modules such as parts inventory, purchasing, vendor management, or asset depreciation unless those are needed later.

Most available systems cover pieces of this workflow. Matching the full combination usually requires multiple products or higher-tier plans.

MaintainX

Monthly: \$20-\$75 per user on published paid tiers.

Startup/equipment: implementation may be separate; phones/tablets, labels, and any kiosk setup are usually customer-side. Strong CMMS, but not this full zoo-specific workflow out of the box.

OrangeQC

Monthly: about \$250-\$500, plus add-ons and extra inspectors.

Startup/equipment: QR shortcuts exist, but field devices, printed codes/tags, and rollout work are still separate. Good inspection/ticket fit, not the whole custom operations hub.

UpKeep

Monthly: about \$24-\$55 per user on published tiers.

Startup/equipment: published implementation/training packages range from about \$500 to \$5,000; field devices/labels are still separate. Broad platform, more than custodial needs.

Why the monthly cost jumps

Commercial tools often become expensive once several paid users, QR workflows, offline mode, analytics, implementation help, request portals, API access, or multiple modules are needed.

Initial setup difference

Many systems do not include phones, tablets, printed QR labels, NFC tags, kiosk setup, data cleanup, or local rollout work in the monthly subscription. Those are separate startup costs.

Most available systems cover pieces of this workflow. Matching the full combination usually requires multiple products or higher-tier plans.

Limble CMMS **Monthly: quote/calculator model on public page.**

Startup/equipment: QR/work request features exist, but pricing and rollout costs depend on quote; devices and labels are typically separate. Strong CMMS, broader than this use case.

SafetyCulture **Monthly: team inspection/checklist platform pricing.**

Startup/equipment: mobile devices, printed QR codes, and rollout configuration are outside the base software concept. Good inspections, but not schedule + messenger + events + Memphis AI.

Is there one system this comprehensive?

Public vendor offerings cover important pieces, but one ready-made system combining zoo custodial scans, guest QR issues, events, schedule reassignment, Messenger, reminders, and Memphis AI is not obvious.

Cost conclusion

The custom path wins on cost because current monthly software cost is \$0, startup hardware is minimal, and future spending is limited to replacement phones, replacement tags, or optional cloud upgrades.

QR reporting, smart restrooms, mobile inspections, work requests, and facility dashboards are already common ideas. The difference here is the combined local workflow.

VERIFIED PUBLIC EXAMPLE

Georgia Tech — Smart Restroom Customer Enhancement Program

- Georgia Tech publicly describes campus-wide Smart Restrooms where users scan QR codes in participating restrooms to report issues.
- Submissions go directly to Building Services for faster response and cleaner, better-maintained restrooms.
- The program is described as nationally recognized with APPA’s Effective & Innovative Practices Award.

BROADER MARKET EVIDENCE

- MaintainX, UpKeep, Limble, OrangeQC, and SafetyCulture all publish features around mobile work, checklists, QR/barcode scanning, reports, request portals, or inspections.
- This confirms the category is real: facilities teams are moving away from paper, radio-only workflows, and delayed inspection visibility.
- What is less common publicly is one lightweight system combining custodial area ownership, NFC work confirmation, guest QR reports, events-by-area notices, due/overdue reminders, internal messaging, and AI answers in one zoo-specific flow.

HOW TO PHRASE THIS SAFELY

The strongest claim is not “no one has anything like this.” The stronger and safer claim is: public products and examples prove the individual ideas are valid, but this custom system combines them around the zoo’s exact custodial operation at a much lower starting cost.

SOURCES CHECKED

Georgia Tech Facilities custodial page; MaintainX pricing; OrangeQC pricing; UpKeep pricing; Limble pricing/features; SafetyCulture pricing; Supabase pricing; Render pricing; GitHub plans documentation.